Analytical Chemistry Cumulative Exam
September 6, 2013

Note: To pass this exam, you have to have to demonstrate sufficient understanding of the concepts in all sub-sections.

I. General Analytical Chemistry Knowledge.

1) What is the pH of an aqueous solution of 1 nM HNO_3? Justify your answer.

2) Give an operational definition for the following terms used in analytical chemistry. Give examples or draw a picture if it helps your argument.

a) Sensitivity b) Selectivity c) Specificity

II. Problem Solving Capability (Short Answer).

State which instrument could be used to solve the problems listed. Provide a terse justification for your choice.

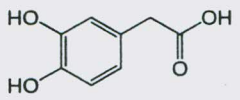
1) Determining whether a chemical modification that would add a group with the element fluorine has occurred onto a material containing only carbon, hydrogen, and oxygen.

2) Measure caffeine content in coffee.

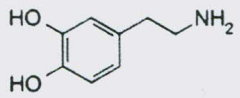
3) Determine the redox properties of a new inorganic catalyst.

III. Problem Solving Capability (Longer Answer).

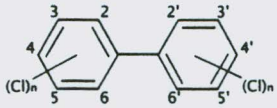
Pick **one** problem from the list below and discuss how this analysis problem could be solved. Be sure to think about sample preparation, instruments to use and data analysis. When discussing the instruments, justify why you want to use that instrument and provide a brief theoretical description of how the instrument works. If there might be more than one way to solve the problem, pick one method to describe in depth and mention alternative methods (single sentences). Be sure to discuss calibration, if needed, and any other potential problems given the problem statement. I will only grade one answer.



DOPAC



Dopamine



PCB General Structure

a) Absolute quantitation of DOPAC and dopamine obtained from biopsied brain tissue. Note, these compounds are quite hydrophilic. Their structures are shown to the side.

b) Measuring the amount of free lead (lead ion, Pb^{2+}) vs. total lead (including bound and organic lead) in fish.

c) Quantifying at least 50 of the 209 possible polychlorinated biphenyls (PCBs) from river sediment samples. Note, these compounds are hydrophobic, but with sufficient temperature (~ 250°C) can be made into a vapor and are stable at this temperature.